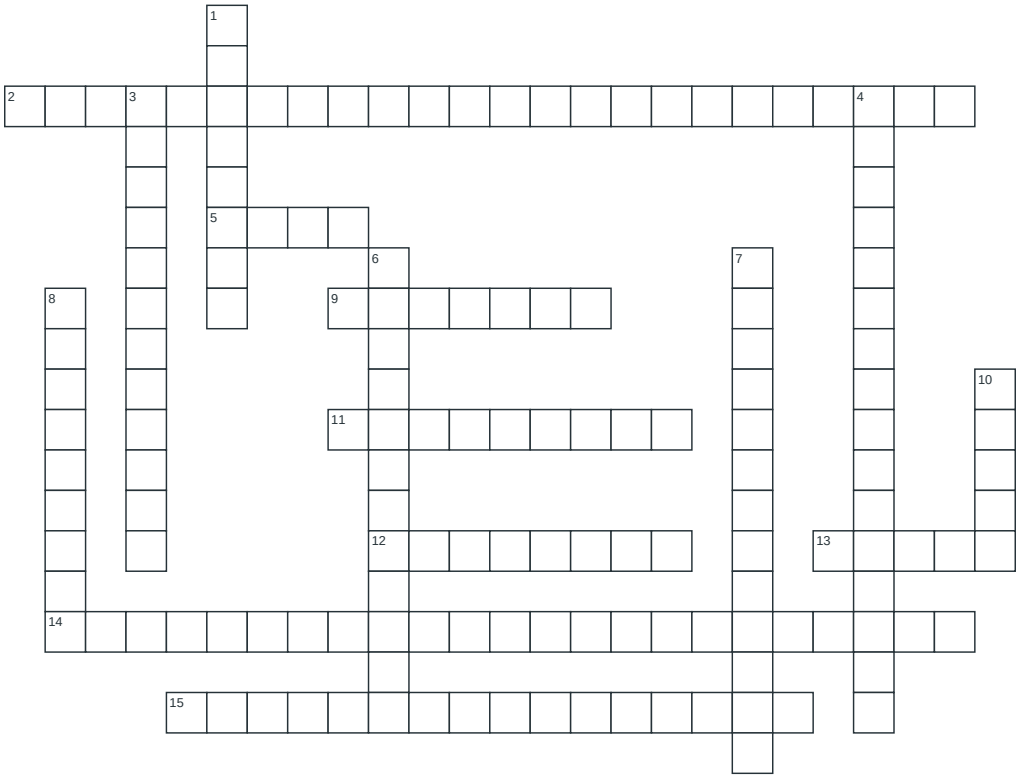
Wave Interference and Resonance Crossword

Medium | 15 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

2. Overlapping displacements reinforce each other
5. A point with zero displacement amplitude
9. Reduces oscillation amplitude over time
11. A strong response when driving frequency is near a system's natural frequency
12. Point of maximum displacement in a standing wave.
13. Single, localized disturbance that travels through a medium.
14. Opposite displacements partly or completely cancel at overlap
15. The combination of overlapping wave displacements

DOWN

1. Frequency that is an integer multiple of a system’s fundamental frequency.
3. Pattern formed by overlapping waves traveling in opposite directions.
4. Frequency at which a system tends to oscillate when disturbed.
6. Reduction in displacement when waves overlap with opposite phases.
7. Rate of loudness variation produced by two nearby sound frequencies.
8. Distance traveled by a wave per unit time.
10. Position within a repeating wave cycle.

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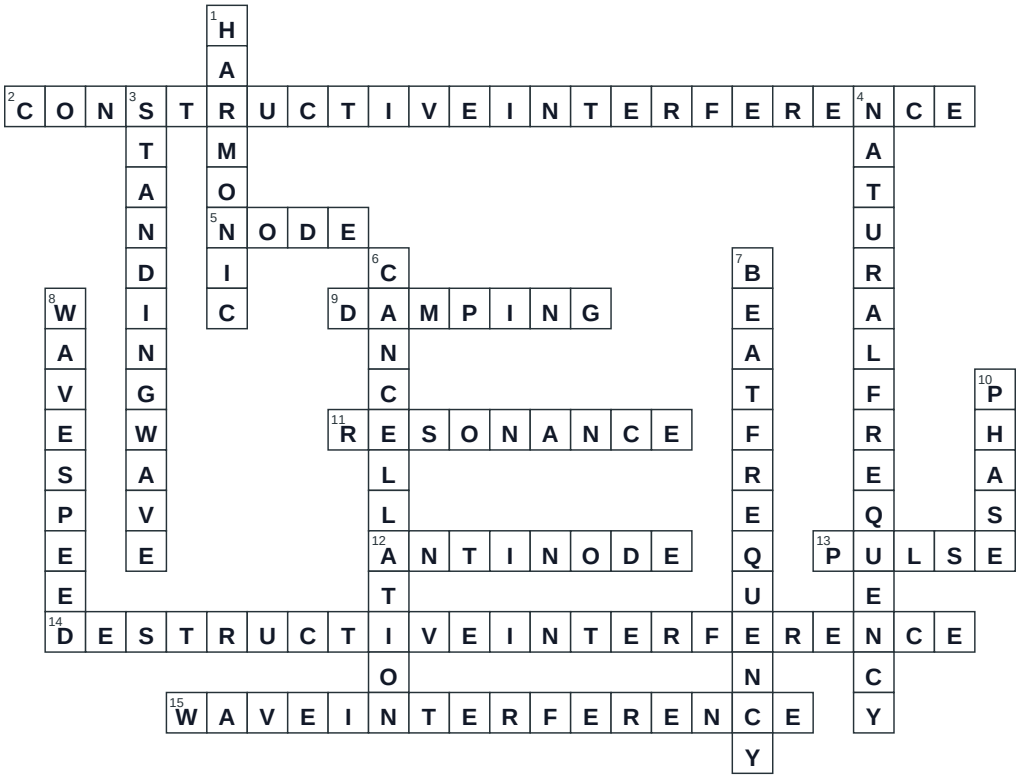
Medium | 15 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

2. Constructive Interference
5. Node
9. Damping
11. Resonance
12. Antinode
13. Pulse
14. Destructive Interference
15. Wave Interference

DOWN

1. Harmonic
3. Standing wave
4. Natural frequency
6. Cancellation
7. Beat frequency
8. Wave speed
10. Phase